FlyCommerce OS: Capstone Portfolio Case Study

Predictive Multi-Channel E-Commerce Fulfillment & Inventory Hub · UX/UI Track · Final Submission

NN/G STRUCTURED CASE STUDY

ROLE	DURATION & TRACK	PRIMARY TOOLING	AUTHOR
Lead Product Designer	8-Week Sprint · UX/UI Design	Figma, Claude 3.5, DeepSeek	K. M. Siam (@KMSiam)

USABILITY SUS SCORE	TIME ON TASK	TASK COMPLETION	ACCESSIBILITY
89.0 (+17.5)	**54 sec** (-72%)	**96.7%**	**100% AAA**
Top 5% percentile (Grade A+ Excellent)	Triage & rebalance dropped from 3m 12s	Verified across 5 operations leads	Lowest body text contrast is 7.3:1

1. The Problem & User Discovery

Direct-to-Consumer (DTC) brands scaling from $2M to $20M GMV across **Shopify, TikTok Shop, and Amazon FBA** suffer from a catastrophic 42-minute batch inventory sync lag. When viral spikes occur on TikTok Shop, inventory runs dry; yet Shopify and Amazon continue accepting orders for phantom stock. This triggers stockouts, customer churn, seller rating bans, and over $180,000 in annual emergency freight penalties.

> *"I don't need a black-box AI that silently re-routes $50,000 of stock behind my back. I need transparent velocity metrics, 60-second reversible safeguards, and touch-friendly actions I can execute on an iPad cart in a loud warehouse."*
> ***— Maya Lin, Head of E-Commerce Supply Chain Operations (Primary Persona)***

2. The Strategic Pivot & IA Flow Architecture

Strategic Pivot ("Kill Your Darlings")

In early wireframes (UX-04A), we designed a complex 3D interactive warehouse bin visualizer. User testing proved operators found it slow, distracting, and unviable under poor warehouse Wi-Fi.

Pivot: Cut the 3D visualizer in favor of a high-density, sub-second data table paired with a 1-click predictive rebalance drawer.

End-to-End User Flow Arc

Trigger: Automated velocity detector flags TikTok spike (6.2h runway remaining).
Inspect: Operator opens AI governor drawer with live formula transparency.
Act: 1-click approval creates warehouse transfer slips with a 60-second undo countdown.

3. Design System Foundations: FlyCore v1.0

Built on Figma Variables across Light and Dark modes. Zero detached components; 100% adherence to scalable production standards:

8pt Geometric Scale	Major Third (1.250)	36 Component Matrix	WCAG 1.4.1 Shapes
4px, 8px, 16px, 24px, 32px, 48px, 64px auto-layouts	Display (32), H1 (24), H2 (20), Body (14), tnum figures	Buttons, inputs, cards, modals across all interaction states	Hexagon (⬡), Triangle (△), Square (□) paired with color

4. Designing the AI Surface: Trust & Human Governance

Shape of AI Pattern	Implementation in FlyCommerce OS	User Trust & Safety Mechanism
Reversible Action Plan (Governor)	Stock transfer drawer displays before/after balances across all channels prior to execution.	60-second visual countdown with instant 1-click cancellation to prevent remorse.
Formula Disclosure (Trust Builder)	Inline card explicating calculation: `(TikTok Velocity × 1.4) - Transit Lag`.	No black-box opacity; operators understand exactly which historical sales window was parsed.
Graceful Degradation (Failure Mode)	Ambiguity banner when marketplace APIs lag or experience rate limits.	System expands safety stock buffers by +15% automatically and flags reduced confidence.

5. Usability Testing & Top 3 Iterations (UX-09)

Moderated think-aloud sessions with 5 enterprise operations leads surfaced 8 defects; top 3 were remediated immediately:

Observed Friction	Severity	Prototype Iteration Shipped	Post-Iteration Outcome
Excess Reserve Rebalance	P0 Critical	Added dynamic depletion gauge; turns amber <15% and locks transfer if buffer hits 0.	0 over-depletions across re-tests.
Hidden Formula Logic	P1 Major	Elevated formula from hover tooltip into a persistent inline card with confidence score.	Hesitation dropped from 48s to 6s.
Warehouse Tablet Mis-Taps	P1 Major	Enlarged action targets to 54px full-width touch areas on warehouse tablet view.	100% first-tap accuracy achieved.

6. AI Workflow Reflection: What Worked, What Failed & What Stayed Human

Where AI Failed (Required Human Fixes)

- **Contrast Hallucinations:** Claude suggested color tokens that failed WCAG AA (4.4:1). Human remediation was required to achieve 7.3:1 AAA.
- **Dashboard Bloat:** AI wireframes defaulted to 14 useless visual charts, missing the core triage workflow.

What Stayed 100% Human

- **Empathy with Operator Panic:** Understanding the terror of Amazon seller rating penalties cannot be simulated.
- **Moderating Usability Nuance:** Detecting subtle sighs, hesitation pauses, and unspoken user uncertainty.

7. Critical Reflections

One Decision I Would Defend

High-Density Data Grid over "Spacious Card Tiles": While modern Dribbble shots favor cards with massive padding, operators triaging 4,000 SKUs need immediate vertical scanability and tabular numbers. This choice cut task time by 72%.

One Thing I Would Do Differently

Simulate Warehouse Environmental Friction Earlier: Testing on quiet desktops masked touch-target and glare issues until Week 6. In the future, I will simulate noisy warehouse conditions and tablet grips during initial low-fi wireframing.

Deliverable Links & Portfolio Verification:

Interactive Prototype: Figma Prototype Link · Master Project File: Figma Design File · GitHub Repository: FlyRank Internship UX/UI Track